

Class nr. 12

Subject: 1. Consumer Theory

1.2. Slutsky Equation (practice classes)

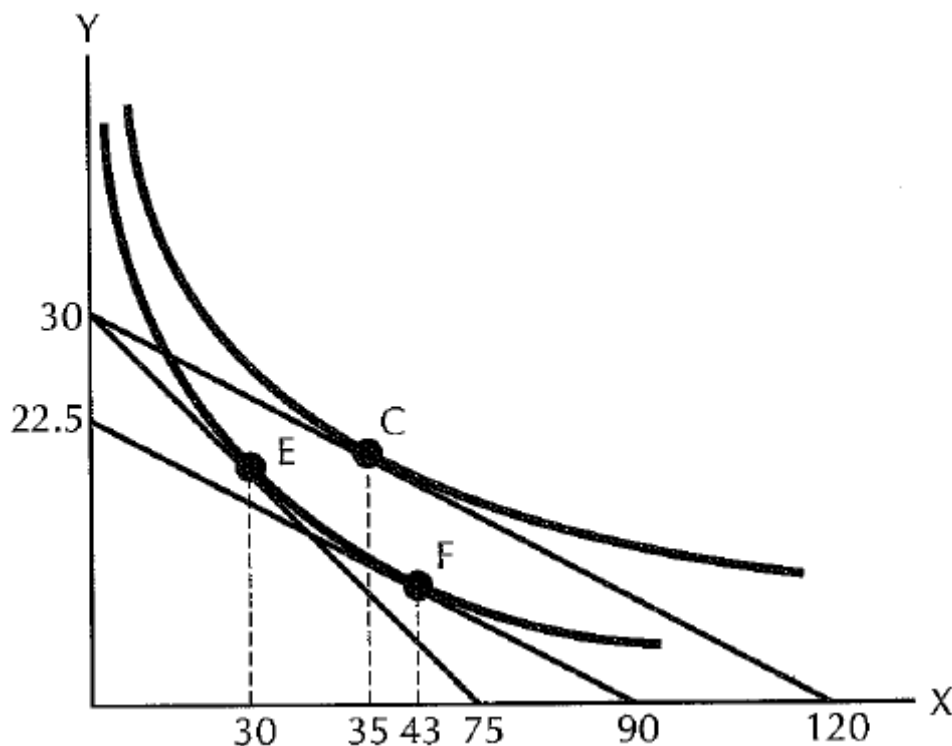
Exercise 8.1. from Bergstrom and Varian's book (2006) "Workouts in Intermediate Microeconomics", p. 96-97

Gentle Charlie, vegetarian that he is, continues to consume apples and bananas. His utility function is $U(x_A, x_B) = x_A x_B$, where x_A is the quantity of consumed *apples* and x_B is the quantity of consumed *bananas*. The price of apples is \$1, the price of bananas is \$2, and Charlie's income is \$40 a day. The price of bananas suddenly falls to \$1.

- a) How many bananas and apples did Charlie consume per day, before the price change? Sketch, in a graph, Charlie's original budget constraint and the chosen consumption bundle.
- b) If after the price change, Charlie's income had changed so that he could exactly afford his old consumption bundle, what would his new income have been? With this income and the new prices, how many apples and bananas would he consume? Sketch, in a graph, Charlie's new budget constraint and the chosen consumption bundle.
- c) Does the substitution effect of the fall in the price of bananas make him buy more bananas or fewer bananas? How many more or fewer?
- d) How many apples and bananas does Charlie actually buy, after the price change? Sketch, in a graph, Charlie's budget constraint and the chosen consumption bundle. Determine and sketch, in a graph, the total, substitution and income effects of the price change on the demand for bananas.
- e) Determine the direction and magnitude of the income change equivalent to the previously determined income effect. Explain the impact on the consumption of bananas measured by the income effect.
- f) Does the substitution effect of the fall in the price of bananas make Charlie consume more *apples* or fewer? How many more or fewer? Does the income effect of the fall in the price of bananas make Charlie consume more apples or fewer? How many more or fewer? What is the total effect of the change in the price of bananas on the demand for apples?

Exercise 8.3. from Bergstrom and Varian's book (2006) "Workouts in Intermediate Microeconomics", pp. 98-99

Consider the figure below, which shows the budget constraint and the indifference curves of King Zog. Zog is in equilibrium with an income of \$300, facing prices $p_X = \$4$ and $p_Y = \$10$.



- How much X does Zog consume?
- If the price of X falls to \$2.50, while the income and the price of Y stay constant, how much X will Zog consume?
- How much income must be taken away from Zog to isolate the Hicksian income and substitution effects?
- Make use of the points C , E and F to indicate the total effect of the price change.
- Make use of the points C , E and F to indicate the substitution and income effect of the price change.
- Is X a normal good or an inferior good?
- Sketch an Engel curve and a demand curve for good X that would be reasonable given the information in the graph above.